Kerr Landau-Siegel proof evaluation, Step 1: structure map and foundation inventory

Document evaluated: Bryce Kerr, "A simpler proof that there are no Landau–Siegel zeros," arXiv:2610.21099v1 [math.NT], 15 October 2026, 6 pages (the copy Arnoldo attached on 18 October 2026).

Standard applied: a proof must carry the reader from step to step with the reader supplying nothing, judged only by what is on the page. Every result taken from outside the paper must be stated up front as foundation.

Scope of this step: a map of what the paper proves, what each part depends on, and every outside result it uses, whether it names that result or not. This step judges no step to be correct or incorrect. Breaks and false steps are catalogued in Step 2. Citations are checked against the originals in Step 4.

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1. Parameters and the order in which they must be chosen

The proof uses the following parameters. The paper never lists them in one place. The order below is what the proof of Theorem 1 needs, read off its final paragraph ("provided γ sufficiently large, ε sufficiently small in terms of γ, and |q| sufficiently large").

Order	Parameter	Definition on the page	Depends on
0	C_1	implied constant in Lemma 1 (absolute)	nothing
1	γ	positive integer, "sufficiently large"	C_1 (implicitly)
2	ε	"sufficiently small in terms of γ"	γ, C_1
3	|q|	"sufficiently large"	γ, ε, and the "sufficiently large N" of Lemma 5
4	N	|q|^γ	γ, q
5	B	8N√|q|	N, q
6	D	⌊½((N − 1)/3)^{4/3}⌋	N
7	Q	D/(log N)²	D, N
8	S	primes p with 8 < p ≤ Q, p ∤ 2q, χ(p) = −1	Q, q
9	k_p	max{1, ⌊D/p⌋}	D, p
10	P, r, E_p, W, H	as in Lemmas 4 and 5; H = P/(4r)	all of the above

The final constant c of Theorem 1 is never named on the page. It must come from ε (for large |q|) together with a separate treatment of small |q|, which the page does not contain. Step 2 records this.

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2. Structure map

2.1 Statements and dependencies

Item	What it states	Proof on the page?	Uses	Used by
Theorem 1	(1 − β) log|q| ≥ c for every real zero β ∈ (0, 1) of L(s, χ), χ = (q/·), q a fundamental discriminant	Yes, by contradiction (pages 5 to 6)	Lemmas 1, 5, 6, 7; F4, F5, F11	(main result)
Lemma 1	$\Sigma_{p\le X,\ \chi(p)=1}(\log p)/p \ll \log\|q\| + (1-\beta)(\log X)^2$, for X ≥ 3	No: called "a straightforward consequence" of F1	F1	Theorem 1
Setup of K, O	K = Q(√q, √2) has degree 4; O = Z[√q, √2] has Z-basis 1, √q, √2, √(2q); four embeddings; norm ‖·‖; automorphisms σ, τ	Stated, not proved	F7	Everything after
Equation (2)	$\theta_n = n_1 + n_2\sqrt{q} + n_3\sqrt{2} + n_4\sqrt{2q}$, with $n \in \{0, \dots, N-1\}^4$	Definition		Lemmas 4, 5; Theorem 1
Equations (3), (4)	For p ∤ 2q with χ(p) = −1: $\theta_n^p \equiv \sigma\theta_n$ or $\sigma\tau\theta_n \pmod{pO}$, according as (2/p) = 1 or −1	Partly: the congruences for √q and √2 are cited; the step to θ_n^p is "Hence"	F2, F3, F9	Lemma 4
Definition of L_D	O-lattice of polynomials Σ	Definition		Lemmas 2 to 5

	$c_{ijk}\, x_1^i\, x_2^j\, x_3^k$ over monomials with $i + 8(j + k) \le D$			
Lemma 2	For $p > 8$ and $\ell \in \{2, 3\}$, the polynomials $G_{ijk} = x_1^v (x_1^p - x_\ell)^e x_2^j x_3^k$ ($i = pe + v$) form a basis of L_D	One sentence: "change of basis matrix ... triangular ... diagonal entries one"	F8	Lemma 4
Definition of $\|f\|_B$	$\max \|c_{ijk}\| B^{i+j+k}$	Definition		Lemmas 3, 5
Lemma 3	$\#\{f \in L_D : \|f\|_B \le H\} > \Pi (H/(8\sqrt{\|q\|} B^{i+j+k}))^4$	Yes, short	Basis of O	Lemma 5
Lemma 4	The number of value vectors $(f(\theta_n, \sigma\theta_n, \sigma\tau\theta_n) \bmod PO)$*n, as f ranges over L_D, is at most $P^{4r} \Pi_{p\in S} p^{-4E_p}$*	Yes	Lemma 2; (3), (4); F6; F7 (size of $O/p^m O$)	Lemma 5
Lemma 5	For N large: if $\Sigma_{p\in S} (\log p)/p > (13/10) \log B$ (9), then a nonzero $F \in L_D$ vanishes at every $(\theta_n, \sigma\theta_n, \sigma\tau\theta_n)$ (10)	Yes, with asymptotics stated without derivation	Lemmas 3, 4; F7 (norm); F10; F12	Theorem 1
Definition of $\mu(\Gamma)$	min over proper subspaces L of C^n of $\text{rank}_Z(\Gamma/(\Gamma \cap L)) / (n - \dim_C L)$	Definition		Lemmas 6, 7
Lemma 6	If a nonzero $F \in C[x_1, \ldots, x_n]$ vanishes at every $\Sigma m_j \gamma_j$ with $0 \le m_j \le M$, then $\deg F \ge (M/n)^{\mu(\Gamma)}$	No: attributed to Masser [5]	F4	Theorem 1
Lemma 7	If $\pi: C^4 \to C^3$ is surjective and $\ker \pi$ lies in no proper rational subspace, then $\mu(\pi(Z^4)) \ge 4/3$	Yes	F8	Theorem 1
Kernel claim (in proof of Theorem 1)	$\det(\gamma_1, \gamma_2, \gamma_3) = 4\sqrt{(2q)} \ne 0$; $\ker \pi$ is spanned by $v = (\sqrt{(2q)}, \sqrt{2}, -\sqrt{q}, -1)$; the coordinates of v are $\mathbb{Q}$-independent, so $\ker \pi$ lies in no proper rational subspace	Determinant and kernel stated; independence asserted	Setup of K; F8	Lemma 7 applied

2.2 Logical chain of the main proof

1. Assume $(1 - \beta) \log|q| < \varepsilon$.
2. Choose γ, then N, B, D, Q, S.
3. Mertens (F5) and Lemma 1 give $\Sigma_{p\in S} (\log p)/p \ge \log Q - O(\log|q| + (1 - \beta)(\log Q)^2)$.
4. Use $\log Q = (4/3) \log N + O(\log \log N)$ and $\log B = \log N + \tfrac{1}{2} \log|q| + O(1)$ to get (9). The working margin is $4/3 - 13/10 = 1/30$ in the coefficient of $\log N$.
5. Lemma 5 gives a nonzero $F \in L_D$ vanishing at all N^4 points.
6. The points are $n_1\gamma_1 + \ldots + n_4\gamma_4$ with $0 \le n_j \le N - 1$. Lemma 6 ($n = 3$, $M = N - 1$) and Lemma 7 give $\deg F \ge ((N - 1)/3)^{4/3}$.
7. $F \in L_D$ forces $\deg F \le D < ((N - 1)/3)^{4/3}$. Contradiction.

2.3 The internal margin chain

Three constants must sit in the order $5/4 < 13/10 < 4/3$:

- **5/4** comes from the pigeonhole in Lemma 5: $W \sim 5D^4/6144$ and $p \cdot E_p \sim D^4/1536$, and the ratio of the leading constants is $(5/6144)/(1/1536) = 5/4$. This is what (11) requires in the limit.
- **13/10** is the threshold the paper chooses in (9). It leaves room of $1/20$ for the error terms in Lemma 5.
- **4/3** is $\log Q / \log N$ in the limit. It comes from $D \asymp N^{4/3}$, which in turn is set by the exponent $\mu \ge 4/3$ that Lemmas 6 and 7 supply.

The multiplicative constant (M/n) in Lemma 6 is not load-bearing. A different constant changes D by a constant factor, which changes $\log Q$ only by $O(1)$. The exponent $\mu(\Gamma)$ **is** load-bearing. An exponent below 13/10 would break the argument.

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3. Foundation inventory

Each item is a result from outside the paper that the argument uses. The column "Stated on the page?" records whether the paper names it as an assumption, cites it, or uses it silently.

#	Result	Where used	Stated on the page?	To be checked in
F1	Davenport, *Multiplicative Number Theory*, 2nd ed., p. 89, Equation (2)	Lemma 1	Cited, with the derivation of Lemma 1 omitted	Step 4 (original); Step 2 (missing derivation)
F2	Frobenius congruences in O: (√q)^p ≡ χ(p)√q and (√2)^p ≡ (2/p)√2 (mod pO), for p ∤ 2q	(3), (4)	Cited: Milne, §8, Example 8.19	Step 4. Note: both follow in one line from Euler's criterion, F3
F3	Euler's criterion; for odd p ∤ q, the Kronecker symbol (q/p) equals the Legendre symbol	F2; definition χ(p) = (q/p)	Silent	Step 2
F4	**Masser's zero estimate**, as stated in Lemma 6	Lemma 6, hence Theorem 1	Cited: Masser, Invent. Math. 63 (1981), 81 to 95	**Step 4. Load-bearing:** the exponent μ(Γ), this definition of μ, the range $0 \le m_j \le M$, and whether Masser states the result in this form
F5	Mertens' estimate: Σ_{p≤Q} (log p)/p = log Q + O(1)	Proof of Theorem 1	Named ("Mertens' estimate") but not stated	Step 2
F6	Chinese remainder theorem for O/PO ≅ Π_p O/p^{k_p}O	Lemma 4	Named, not stated	Step 2
F7	Facts about K and O: [K : Q] = 4 (requires q ≠ 8 and q not a square); O is free with basis 1, √q, √2, √(2q); |O/p^m O| = p^{4m}; K has four embeddings and Galois group {1, σ, τ, στ}; the norm N_{K/Q} maps O ∖ {0} to nonzero integers and is multiplicative; |N(α)| ≤ ‖α‖⁴	Setup; Lemmas 4, 5; kernel claim	Asserted ("The field K has degree four") or silent	Step 2. The degree claim is false as stated for q = 8
F8	Linear algebra: a unitriangular integer change of basis is invertible over Z (hence over O); a subspace of C^n spanned by rational vectors is cut out by rational linear forms; the rank of a subgroup of Z^4 equals the complex dimension of its span; dim π(W) = dim W when ker π ∩ W = {0}	Lemma 2; kernel claim; Lemma 7	Silent	Step 2
F9	Binomial theorem modulo p in O, and n^p ≡ n (mod p) for integers n (Fermat)	(3), (4)	Silent	Step 2
F10	Pigeonhole principle	Lemma 5	Named	None needed
F11	Chebyshev bound: Σ_{p≤Q} log p ≪ Q	Lemma 5 (the $O(D^3Q)$ error term)	Silent	Step 2
F12	Properties of the embedding norm ‖·‖: triangle inequality, submultiplicativity, ‖σα‖ = ‖τα‖ = ‖α‖	Lemmas 3, 5	Silent	Step 2

Convention never stated: to apply Lemma 6, the points (θ_n, σθ_n, στθ_n) and the vectors $\gamma_1, \ldots, \gamma_4$ must lie in C^3. That requires fixing one embedding K ↪ C, and fixing √(2q) = √2 · √q. For q < 0 the branch of √q must also be fixed. The paper does not say this. It matters for the kernel vector v and for $\det(\gamma_1, \gamma_2, \gamma_3) = 4\sqrt{(2q)}$.

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4. Observations from this step (structure only; no verdicts)

1. **Two results carry the proof from outside:** Lemma 1 (F1) and Lemma 6 (F4). Everything else is either proved on the page or is standard algebra and elementary prime-number estimates.
2. **Lemma 6 is the decisive import.** Its exponent fixes the 4/3 at the top of the margin chain (Section 2.3). The paper gives no statement of where in Masser's paper the result appears (no theorem number), so even locating it is left to the reader.
3. **The paper does not state its foundation up front.** The "Analytic input" paragraph introduces Lemma 1, and Lemma 6 is introduced at the point of use. Under our standard this is the first break; Step 2 will record it with the others.
4. **The conclusion is incomplete as a structure.** The proof by contradiction covers only |q| large, depending on γ and ε. No sentence covers the remaining finitely many q or names c.